



PROJECT-BASED LEARNING

PHASE-I EVALUATION

SmartFlow: Adaptive Traffic Signal Management System

Team ID: T168

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: *T168*
Semester: *3rd*
Section: *L, E, ML-1, F*
Domain: *Data Structures & Algorithms*

Team Members

1. *Dipansh – 2510019668*
2. *Anshuman Negi – 2510012036*
3. *Nidhi Bhatt – 2510370049*
4. *Aditya Nautiyal – 2510011111*

Mentor

Dr. Narayan Chaturvedi

Interactions completed: *3*

Problem Statement & Motivation

Problem Statement

Current traffic signals operate on fixed timings, which often lead to congestion and increased waiting times during varying traffic conditions. This project proposes a traffic simulation system that analyzes vehicle density and dynamically adjusts signal timings to improve traffic flow, reduce delays, and enhance overall road efficiency.

Motivation

- ▶ *Fixed-time traffic signals do not adapt to traffic density.*
- ▶ *Increased waiting time and congestion.*
- ▶ *Advanced adaptive systems are expensive.*

Target Users / Stakeholders

- ▶ *Commuters and Drivers*
- ▶ *Traffic Management Authorities*
- ▶ *Reduced waiting time and improved traffic flow.*

Objectives

Project Objectives

1. *To simulate a four-ways intersections using queue.*
2. *To develop a density-based traffic signal allocation system.*
3. *To evaluate performance using average vehicle waiting time.*
4. *To implement concepts of Object-oriented programming*
5. *To implement concepts of Data Structures & algorithm*

Key Objective: *To optimize traffic flow by dynamically allocating signal priority.*

Proposed Solution

Proposed Approach

1. *Study the traffic signal problem and identify requirements.*
2. *Design the queue-based traffic management architecture.*
3. *Implement vehicle generation and lane queue modules.*
4. *Develop the density-based signal allocation algorithm.*
5. *Test and evaluate system performance.*

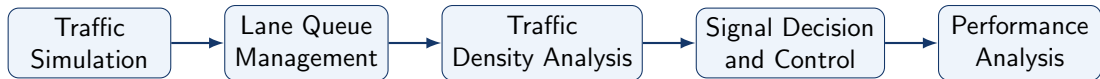
Key Features

- ▶ *Adaptive Signal Timing.*
- ▶ *Traffic density analysis.*
- ▶ *Fair signal allocation.*
- ▶ *Fixed timer comparison.*
- ▶ *Performance analysis.*

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	<i>Queue, Arrays, Searching, ADTs.</i>	<i>Traffic Lane Management</i>
OOPs with C++	<i>Classes, Object, Encapsulation, STL.</i>	<i>System Architecture & Simulation</i>

System Architecture / Workflow



Major Components

- ▶ *Traffic Simulation Module*
- ▶ *Queue Management Module*
- ▶ *Decision-Making Module*
- ▶ *Signal Control & Analysis Module*

Data / Control Flow

- ▶ *Vehicles are generated and stored in lane queues.*
- ▶ *Traffic density is analyzed and signal priority is determined.*
- ▶ *Signals are allocated and system performance is evaluated.*

Technology Stack & Methodology

Technology Stack

- ▶ Programming: *C++*
- ▶ Tools / Framework: *STL, Overleaf*
- ▶ IDE: *Virtual-Studio Code*
- ▶ Version Control: *Git / GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

Initial Progress & Team Contribution

Progress Achieved

- ▶ *Problem Statement and Project Objectives Finalized*
- ▶ *Study of Existing Traffic Management Systems Completed*
- ▶ *Technology Stack Finalized*
- ▶ *System Architecture Designed*

Individual Contribution

Member	Role	Contribution
Dipansh	Lead	System Architecture Design
Anshuman	Developer	Problem Analysis & Objectives
Nidhi	DB/Testing	Technology Stack & Development Planning
Aditya	Developer/Docs	Domain Research & Documentation

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation
4. Phase-IV: Testing & Demonstration

Expected Outcomes

- ▶ *Scalable Smart Traffic Solution*
- ▶ *Reduced Average Waiting Time*
- ▶ *Performance Comparison with Fixed-time signal system*

References

1. *Balagurusamy (2020) – OOP with C++*
2. *Cormen et al. (2022) – Introduction to Algorithms*
3. *Goodrich et al. (2011) – DSA in C++*
4. *Stroustrup (2014) – Programming: Principles and Practice Using C++*

Thank You
Questions & Suggestions